

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NADH Dehydrogenase Complex Assembly (GO:0000000)	-0.25231564	41	2.340e-08	6.285e-05	TMEM126A:20 TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 ECSIT:223 NDUF9A:353
Mitochondrial Respiratory Chain Complex	-0.25231564	41	2.340e-08	6.285e-05	TMEM126A:20 TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 ECSIT:223 NDUF9A:353
Mitochondrial Respiratory Chain Complex	-0.19480931	62	1.179e-07	2.110e-04	TMEM126A:20 TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 ECSIT:223 NDUF9A:353
Mitochondrial Gene Expression (GO:014005)	-0.15155407	91	6.201e-07	8.326e-04	PTCD3:27 MRPL37:33 MTF12:175 MRPL43:176 RARS:226 MRPS30:226 FASKT5:290 LARS:390
Mitochondrial Translation (GO:0032543)	-0.15155407	87	1.034e-06	1.111e-03	PTCD3:27 MRPL37:33 MTF12:175 MRPL43:176 RARS:226 MRPS30:226 LARS:390
tRNA Methylation (GO:0030488)	-0.21803620	34	1.101e-05	9.856e-03	TRMT9B:145 TRMT10C:184 METTL8:444 THUMP02:483 TWY5:157 TRMT10B:598 METTL1:609
Oxidative Phosphorylation (GO:0006119)	-0.18085227	41	6.255e-05	4.799e-02	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 ATP5B:368 TFAM:423 NDUF84:424
Mitochondrial Electron Transport, NADH T	-0.22550045	25	9.591e-05	6.439e-02	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 NDUF84:424 NDUF8A:464 NDUF86:623
Aerobic Respiration (GO:0009060)	-0.17835275	39	1.178e-04	7.033e-02	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 TFAM:423 NDUF84:424 NDUF8A:464
Cell Junction Assembly (GO:0034329)	0.13896227	63	1.398e-04	7.510e-02	SHANK3:7 SHANK2:62 SDK1:143 ARVCF:147 DBNL:169 SKD2:368 POU4F1:379
Calcium Ion Transport (GO:0006816)	0.11348239	93	1.613e-04	7.875e-02	ITPR1:24 TRPM5:49 TRPM1:107 CACNA1:501 CACNG3:266 CRACR2B:285 TRPV6:300
Proton Motive Force-Driven Mitochondrial	-0.17965487	36	1.937e-04	8.670e-02	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 ATP5B:368 NDUF84:424 NDUF8A:464
tRNA Modification (GO:0006400)	-0.13657067	61	2.301e-04	9.508e-02	ALKBH8:116 TRMT9B:145 METTL8:444 THUMP02:483 DTWD1:504 TWY5:157 OSGEP1:1562
Proton Motive Force-Driven ATP Synthesis	-0.17174606	38	2.520e-04	9.666e-02	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 ATP5B:368 NDUF84:424 NDUF8A:464
Cilium Assembly (GO:0060271)	-0.07339778	197	4.111e-04	1.472e-01	ARL13B:21 CEP126:24 FAM161A:31 TCTN3:40 GORA6:68 NDMBP:78 CDC66:105
Calcium Ion Import Across Plasma Membran	0.17509270	33	3.046e-04	1.694e-01	CACNA1B:47 SCN2A:81 TRPM1:107 CACNA1S:201 TRPV6:300 CXCL396 SCN8A:663
mRNA 3'-End Processing (GO:0031124)	0.18425382	28	7.449e-04	2.278e-01	SLBP:3 CSTF2:14 CDC73:73 WDR33:304 SYMPK:386 PCU2:508 CPSF1:704
Muscle Contraction (GO:0006936)	0.11492762	72	7.635e-04	2.278e-01	CACNA1S:201 LMOD1:245 MYH2:322 EMD:399 MYH13:819 RYR2:833 MYOM2:857
Aerobic Electron Transport Chain (GO:0010000)	-0.14145558	43	1.345e-03	2.535e-01	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 TFAM:423 NDUF84:424 NDUF8A:464
Cell-Cell Junction Assembly (GO:0007043)	0.12854013	53	1.226e-03	2.535e-01	ARVCF:147 JAM3:275 TJP1:478 TNL1:514 G3J4:584 TBCD:751 CDH6:781
Cellular Respiration (GO:0045333)	-0.12947663	53	1.128e-03	2.535e-01	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 TFAM:423 NDUF84:424 NDUF8A:464
Cilium Organization (GO:0044782)	-0.06895498	183	1.369e-03	2.535e-01	ARL13B:21 CEP126:24 FAM161A:31 TCTN3:40 DNMBP:78 CDC66:105 KIF21:127
Glycogen Metabolic Process (GO:0005977)	0.29395292	15	1.335e-03	2.535e-01	PYGB:101 PER2:268 PYGM:597 G1Y2:1104 GYG2:1104 GYS2:1280 NHCRL1:1371
Mitochondrial ATP Synthesis Coupled Elec	-0.14145558	43	1.345e-03	2.535e-01	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 TFAM:423 NDUF84:424 NDUF8A:464
Mitochondrial RNA Processing (GO:0000963)	-0.16331748	9	1.018e-03	2.535e-01	SUPV3L1:46 TRMT10C:184 FASKT5:290 PNPT1:511 FASKT2:1042 FASKT4:1105 TBGR4:6080
Negative Regulation Of Fibroblast Prolif	0.28705543	11	9.806e-04	2.535e-01	CDC73:73 MYC1:388 EMD:399 PEK2:508 TRM32:1520 SKI1905 MORC3:2263
Regulative Regulation Of Transcription By	0.04599413	440	1.090e-04	2.535e-01	ZNF281:6 EMT3:33 ZBTB45:39 TCF25:54 TCF15:59 CDC73:73 MXL3
Positive Regulation Of Cytosolic Calcium	0.20811386	20	2.799e-03	2.535e-01	OPRL1:112 F2RL1:217 GPR55:308 GRM1:145 GPR43:466 GPR01:510 P2RY10:727
tRNA Wobble Base Modification (GO:000209	-0.27510634	12	9.699e-04	2.535e-01	ALKBH8:116 TRMT9B:145 ELP2:182 ELPS:815 ELPS:886 MTR01:1010 ELP2:182
DNA Repair (GO:0006281)	-0.05992742	234	1.683e-03	3.013e-01	BRCA1:16 RNF169:49 FANCM:81 RNF168:115 DCLRE1A:125 HROB:314 MH3:147
Chemical Synaptic Transmission (GO:00072	0.06556424	188	2.040e-03	3.402e-01	CACNA1B:47 RIMBP2:76 GABRB3:78 KCNQ3:172 TH:254 RPS6KA2:299 NSG1:314
mRNA Export From Nucleus (GO:0006406)	0.14680814	37	2.018e-03	3.402e-01	MCM3AP:13 IWS1:230 NUP160:372 NUP133:442 YCRF5:192 THOC5:632 DCH3:14
Regulation Of Cilium Movement (GO:000335	-0.33490171	7	2.154e-03	3.402e-01	DNAH11:203 CFAP206:361 CFAP43:704 CYB5D1:725 ALDRE5:220 HCCD5:400 CFAP298:6088
Synaptic Vesicle Uncoating (GO:0016191)	0.43446531	4	2.129e-03	3.402e-01	GAKY:86 SHGSL2:498 SYNJ1:1067 SHG3L3:1301 NA NA NA
Regulation Of Long-Term Synaptic Potenti	0.20185110	19	2.328e-03	3.573e-01	SHANK3:7 ADCY1:25 NSG1:314 SHISA7:1002 YTHDF1:3146 ADCY8:137 SCGN:1558
poly(A)+mRNA Export From Nucleus (GO:00	0.172720566	10	2.406e-03	3.589e-01	MCM3AP:13 IWS1:230 NUP133:442 YCRF5:192 THOC5:632 DCH3:14
mRNA Export From Nucleus (GO:0006405)	0.14017085	38	2.814e-03	3.637e-01	MCM3AP:13 NUP160:372 NUP133:442 ZCYH5:128 NUP155:633 THOC5:632
Adenylyate Cyclase-Inhibiting G Protein-C	0.14121403	37	2.979e-03	3.637e-01	OPRL1:112 GNAI3:239 HRH3:356 OPRL1:625 PDE2A:717 DRD2:1061 MCHR1:1103
Adherens Junction Organization (GO:00343	0.16239062	28	2.956e-03	3.637e-01	JAM3:275 INAVA:369 TJP1:478 MTSS1:583 TBCD:751 CDH6:781 TGA4:1041
Cell-Cell Junction Organization (GO:0045	0.12348637	49	2.819e-03	3.637e-01	TGFB1:89 ARVCF:147 F2RL1:217 TJP1:478 TNL1:514 G3J4:584 CDH6:781

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_T22_AMPLICON	0.30081687	32	3.934e-09	2.548e-05	INTS1:71 ADAP1:154 EIF3B:181 TTYH3:249 GRIFIN:301 MAFK:353 FAM20C:454
WP_MITOCHONDRIAL_COMPLEX_I_ASSEMBLY_MODE	-0.25124301	38	8.478e-08	2.745e-04	TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 ECSIT:223 NDUF9A:353
BENPORATH_ES_WITH_H3K27ME3	0.05798049	733	1.329e-07	2.870e-04	RGS9:26 FRMPD4:42 CACNA1B:47 MPM2:48 NDUF85:158 NDUF85:158 NDUF85:158
NIKOLSKY_BREAST_CANCER_T22Q4_AMPLICON	0.39551650	14	3.001e-07	4.169e-04	EP400:1 P2RX2:178 POLE:214 GOLGA3:378 AKK1:425 ULK1:578 DDX1:603
REACTOME_COMPLEX_I_BIOGENESIS	-0.23659871	39	3.219e-07	4.169e-04	TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 ECSIT:223 NDUF9A:353
REACTOME_DEVELOPMENTAL_BIOLOGY	0.05700612	679	5.420e-07	1.580e-04	MPM2:48 ARHGAP39:52 PTK2:63 TRIO:69 ABLM2:72 SCN2A:81 TGFBI:89
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.15805656	79	1.230e-06	1.138e-03	LRPPRC:42 TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353
PEREZ_TP53_TARGETS	0.05013360	813	1.694e-06	1.371e-03	TGFB3R1:4 PHRF1:8 SEC14L5:15 MICAL3:18 BRD1:19 ADCY1:25 APC2:36
NIKOLSKY_BREAST_CANCER_5P15_AMPLICON	0.30743973	20	1.947e-06	1.401e-03	EXOC3:61 CEP72:191 LRRC14B:566 IRX4:621 BRD9:741 SLC6A18:884 TRIP13:897
REACTOME_INFECTIOUS_DISEASE	0.05854877	555	3.013e-06	1.774e-03	ZDHHC8:5 ENTDP1:22 TPR1:24 ADCY1:25 PTK2:63 NUP37:68 CTD1:74
REACTOME_RESPIRATORY_ELECTRON_TRANSPORT	-0.16680093	66	2.848e-06	1.774e-03	LRPPRC:42 TMEM126B:55 NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353
JOHNSTONE_PARVB_TARGETS_3_DN	-0.05761698	568	3.374e-06	1.821e-03	RTKN2:59 PFKFB3:111 ARHGAP11A:114 ZWINT:118 SPAG5:122 LRR1:142 ADY51A1:170
REACTOME_NEURONAL_SYSTEM	0.08110099	270	4.978e-06	2.480e-03	ADCY1:25 CACNA1B:47 SHANK2:62 GABRB3:78 DBNL:169 KCNQ3:172 ADCY9:179
GINESTIER_BREAST_CANCER_ZNF217_AMPLIFIED	0.07995508	253	1.299e-05	6.009e-03	TRAF2:2 BRD1:19 INTS1:71 SPON2:84 EPN1:140 SH3BP2:145 EIF3B:181
WP_2Q37_COPY_NUMBER_VARIATION_SYNDROME	0.13416109	86	1.749e-05	7.550e-03	KIF1A:31 KLHL30:57 TRAF3P1:106 DGKD:115 GPC1:116 MROH2A:126 CAPN10:177
MOOTHA_HUMAN_MITODB_6_2002	-0.06851299	328	2.225e-05	9.005e-03	CTP2:9 SUPV3L1:46 ACADM:57 NDUF3:77 NDUF8B:95 MLH3:147 NDUF85:158
REACTOME_MITOCHONDRIAL_TRANSLATION	-0.13495764	82	2.451e-05	9.336e-03	PTCD3:27 MRPL37:33 MTF12:175 MRPL43:176 MRPS30:226 ERAL1:378 MRPL22:45
GEORGES_TARGETS_OF_MIR192_AND_MIR215	-0.04845997	655	2.908e-05	1.046e-02	WDCP:8 BRCA1:16 ATAD5:36 NIN:38 DST:54 RTKN2:59 FANCM:81
WP_OXIDATIVE_OF_MISPHORYLATION	-0.20296600	35	3.274e-05	1.116e-02	NDUF3:77 NDUF8B:95 NDUF85:158 NDUF9A:353 ATP5B:368 NDUF84:424 NDUF8A:464
MIKELSEN_MEF_HCP_WITH_H3K27ME3	0.05814282	435	3.684e-05	1.142e-02	CADM3:32 ADGBR1:53 OTUD1:245 MAPKBIP2:111 KCNQ3:172 CNTN4:175 SLC04C1:182
REACTOME_TRANSPORT_OF_MATURE_TRANSCRIPT	0.15809903	57	3.705e-05	1.142e-02	SLBP:3 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372 SYMPK:386
NIKOLSKY_BREAST_CANCER_16P13_AMPLICON	0.13206946	78	5.633e-05	1.658e-02	CRAMP1:80 UNKL1:32 JPT2:267 CAPN15:328 RPLUSD:391 MEIOB:419 HBQ1:598
MARTENS_TRETINOIN_RESPONSE_UP	0.05800662	558	6.194e-05	1.744e-02	KIF1A:31 GJB4:34 SHANK2:62 SPON2:84 WKM2:97 CHST15:125 C16orf90:152
MIKELSEN_NPC_ICP_WITH_H3K4ME3	-0.06950007	311	7.039e-05	1.850e-02	WDCP:8 KLHL5:23 A2M:25 ALKBH8:116 DCLRE1A:125 RBMS2:135 MDH1B:178
REACTOME_PROCESSING_OF_CAPPED_INTRON_CON	0.08860110	169	7.141e-05	1.850e-02	SLBP:3 CSTF2:14 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372
REACTOME_DISEASES_OF_DNA_REPAIR	-0.16652387	47	7.833e-05	1.951e-02	BRCA1:16 RAD51AP1:170 ATM:244 MUTHY:362 XRCXC:364 WRN:438
MEISSNER_NPC_HCP_WITH_H3K4ME2_AND_H3K27M	0.07465928	230	1.043e-04	2.501e-02	CADM3:32 ADGBR1:53 STUM:60 PTGFRN:88 TGFBI:89 G3J4:584 SLC04C1:182
MIKELSEN_NPC_HCP_WITH_H3K27ME3	0.07268223	241	1.087e-04	2.513e-02	CADM3:32 ADGBR1:53 SLC04C1:182 CP17:881 TRAF2D:236 KCNQ2:343 V5X1:348
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.05711356	389	1.228e-04	2.743e-02	MPM2:48 ARHGAP39:52 PTK2:63 TRIO:69 ABLM2:72 SCN2A:81 GPC1:116
REACTOME_SIGNALING_BY_RECEPTOR_TYROSINE	0.105789260	376	1.285e-04	2.774e-02	ITPR1:24 PTK2:63 GABRB3:78 EPN1:140 PTPN6:141 ADAP1:154 COL6A1:165
NIKOLSKY_BREAST_CANCER_ZQ012_Q13_AMPLICO	0.11460567	100	1.385e-04	2.894e-02	TCFL5:59 OPRL1:112 CABLES2:137 ZBTB46:186 KCNQ1:212 SYCP2:229 KCNQ2:343
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.19172434	37	1.925e-04	3.896e-02	SLBP:3 CSTF2:14 WDR33:304 SYMPK:386 SLU7:472 ALYRF:592 THOC5:634
REACTOME_TRANSPORT_OF_MATURE_MRNAS_DERIV	0.10912292	31	2.184e-04	4.286e-02	SLBP:3 NUP37:68 NUP210:164 WDR33:304 NUP43:351 NUP160:372
BENPORATH_EED_TARGETS	-0.04049020	676	2.389e-04	4.421e-02	PARP2:123 FRMPD4:42 CACNA1B:47 COL12A1:75 DGK1:95 ZCCHC108 PLXNC1:149
EPERTT_PROGENTOR	0.10402122	105	2.364e-04	4.421e-02	LRPPRC:42 SORD:97 ZWINT:118 HAUS6:289 DCL1:350 NET1:391 WRN:438
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	-0.09957657	113	2.623e-04	4.719e-02	BRCA1:16 RNF168:115 RAD51AP1:170 POLQ:213 MUS81:243 ATM:244 XRCXC:364
MOOTHA_MITOCHONDRIA	-0.05747165	337	3.125e-04	5.470e-02	SUPV3L1:46 ACADM:57 MTT1:171 NDUF3:77 NDUF8B:95 NDUF85:158 MTF12:175
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	-0.04797798	98	3.443e-04	5.717e-02	BRCA1:16 FANCM:81 FANCE:211 ATM:244 MUTHY:362 XRCXC:364 WRN:438
WONG_MITOCHONDRIA_GENE_MODULE	-0.08469153	151	3.397e-04	5.717e-02	LRPPRC:42 NDUF3:77 NDUF8B:95 NDUF85:158 MRPS30:226 NDUF9A:353 ATP5B:368
KEGG_CALCIIUM_SIGNALING_PATHWAY	0.08222773	152	4.828e-04	7.817e-02	ITPR1:24 ADCY1:25 CACNA1B:47 P2RX2:178 ADCY9:179 PTK2B:193 CACNA1S:201

DisGeNET Top pathways by non-permutation

Autism spectrum disorders	0.05744652	369	1.749e-04	4.271e-01	SHANK3:7 HERC2:21 ITPR1:24 EHM1:33 SHANK2:62 JAKM1P:167 TRIO:69
Schizophrenia	0.03284286	1277	1.431e-04	4.271e-01	ZDHHC8:5 SHANK3:7 BRD1:19 HERC2:21 ADCY1:25 RGS9:26 CADM3:32
Cataract, Pulverulent	0.43837711	5	6.871e-04	9.565e-01	GJ3A1:144 BFP52:205 GJAB:635 CRYBB1:900 HSDFA:1458 NA NA NA
Acute necrotizing encephalopathy	-0.25728441	16	3.679e-04	9.565e-01	TMEM126B:55 NDUF3:77 FOXRED:1480 TMMD01:591 NDUF1A:1642 NDUF9A:695 NDUF9A:695
Genu recurvatum	0.27460241	13	6.093e-04	9.565e-01	CTDP1:74 COL1A2:462 CANT1:820 FBN1:882 COL5A1:2141 XYLT1:1191 COMP:1604
Horizontal eyebrow	0.36758899	7	7.580e-04	9.565e-01	DEAF1:849 RERE:956 EBF3:1474 PRDM16:1571 SKI1:1905 TRAPPC9:2338 GABRD:2353
Hypercholesteric kidneys	-0.34351588	5	7.661e-04	9.565e-01	SARS2:19 CEP290:328 TCTN2:503 INVS:1540 RERN:1864 NA NA NA
Malignant neoplasm of fallopian tube	-0.35300461	8	5.458e-04	9.565e-01	BRCA1:16 CASC1:241 ATM:244 NBR1:499 BARD1:719 BRCA2:1199 TP53:6080
NADH:Q(1) Oxidoreductase deficiency	-0.12186539	22	5.849e-04	9.565e-01	TMEM126B:55 NDUF3:77 FOXRED:1480 TMMD01:591 NDUF1A:1642 NDUF9A:695 NDUF9B:1152
Pallor of optic disc	-0.14312575	46	7.938e-04	9.565e-01	TMEM126A:20 ARL13B:21 FAM161A:31 TMEM126B:55 NDUF3:77 CERKL:323 PEK1:432
Progressive macrocephaly	-0.23926967	19	3.070e-04	9.565e-01	TMEM126B:55 NDUF3:77 FOXRED:1480 TMMD01:591 NDUF1A:1642 NDUF9A:695 NDUF9A:695
Aneurysm, Ruptured	0.36124569	7	9.344e-04	6.519e-01	RNH1:231 PPP1R13L:644 NOS3:646 SHC3:959 NIN1:1239 ACE1:1710 GDF1:6651
Central neuroblastoma	0.02983452	1085	1.290e-03	7.414e-01	HERC2:21 ITPR1:24 KIF1A:31 ZNF236:41 MPM2:48 PTK2:63 TGFBI:89
Huntington Disease	0.04723750	403	1.285e-03	7.414e-01	FAM193A:20 ITPR1:24 APC2:36 MPM2:48 TGFBI:89 HSFI:138 SH3BP2:145
Slender finger	0.29552897	10	1.214e-03	7.414e-01	COL12A1:75 COL6A1:165 COL6A2:255 ZNF592:969 LFGN:1177 KANSL1:1448 COL6A3:1619
Non-obstructive azoospermia	0.12898483	51	1.470e-03	7.977e-01	ENTPD6:29 BOLL:206 WDR11:295 DAZL:359 KITLG:395 MEIOB:419 CCNA1:626
CATARACT, COPOCK-LIKE	0.36324971	5	4.910e-03	8.476e-01	GJ3A1:144 BFP52:205 GJAB:635 CRYBB1:900 ISG20:6651 NA NA
COLORBLINDNESS, PARTIAL, DEUTAN SEE	0.17540501	24	2.948e-03	8.476e-01	TGFBI:89 ANXA6:237 LMOD1:245 PLXNA4:630 CCR5:667 OPN1MW:1091 ROKT2:1423
Epilepsy, Temporal Lobe	0.07238175	122	5.910e-03	8.476e-01	GABRB3:78 SCN2A:81 LETM1:150 KCNQ3:317 BIN1:207 SOD1:302 ESRR1:317
Hyperkinesia, Generalized	0.18619478	21	3.151e-03	8.476e-01	ADCY1:25 TH254 CNTNAP2:528 NGF:564 CHRR1:1010 DRD2:1601 ADCY8:1347
Psychoses, Drug	0.22233340	15	2.877e-03	8.476e-01	RGS9:26 ESRR1:317 HTR1A:1265 SLG6A3:1411 SOD2:1584 ACE1:1710 CAMK2B:1770
Psychoses, Substance-Induced	0.25400445	12	2.318e-03	8.476e-01	RGS9:26 ESRR1:317 HTR1A:1265 SLG6A3:1411 SOD2:1584 ACE1:1710 CAMK2B:1770
Abnormal heart beat	-0.10316780	60	6.780e-03	8.476e-01	CPT2:9 PHYH:71 PEK6:88 SLC19A2:236 SYNE2:296 LARS2:390 TS2R:402
Abnormal mitochondria in muscle tissue	-0.19192911	22	1.840e-03	8.476e-01	TMEM126B:55 NDUF3:77 FOXRED:1480 TMMD01:591 NDUF1A:1642 NDUF9A:695 NDUF9B:1152
Attention deficit hyperactivity disorder	0.05216824	287	2.544e-03	8.476e-01	SHANK3:7 HERC2:21 TRIO:69 MC4R:79 ARVCF:147 SORCS2:163 TGFBI:89
Autistic Disorder	0.03805871	482	4.726e-03	8.476e-01	SHANK3:7 EHM1:33 FRMPD4:52 SHANK2:62 JAKM1P:167 GABRB3:78 SCG2A:81
Bleeding tendency	-0.11883687	46	5.343e-03	8.476e-01	THPO:166 MPL:258 KAT6B:514 HBAKE2:519 F7:718 AZM1L:767 ACNS9:17
Carcinogenesis	0.01863596	2559	4.702e-03	8.476e-01	EP400:1 TRAF2:22 PHRF1:8 NSD2:12 CSF2:14 APC2:36 MPM2:48
Cardiac conduction abnormality	-0.10191300	61	5.979e-03	8.476e-01	CPT2:9 PHYH:71 PEK6:88 SLC19A2:236 SYNE2:296 LARS2:390 TS2R:402
Cataionia	0.25770672	6	2.773e-03	8.476e-01	MCL1:294 CNP:529 SLC30A4:786 DIL1:1118 PANX2:1631 PRODH:6651 NA
Chromosome 2q37 deletion syndrome	0.35442964	5	6.059e-03	8.476e-01	HADC4:364 PASK:413 GPR35:465 HDLBP:1206 FARP2:6651 NA NA
Comatose	-0.10653395	62	3.769e-03	8.476e-01	CPT2:9 TMEM126B:55 ACADM:57 NDUF3:77 HMGCL:421 FOXRED:1480 TMMD01:591
Common acute lymphoblastic leukemia	0.20213178	19	2.403e-03	8.476e-01	TMED2:243 KRT20:443 GJA8:635 CD38:744 MME:762 PDX1:1166 DLL1:1174
Cystic renal dysplasia	-0.30542902	9	1.674e-03	8.476e-01	NEK8:245 CEP290:328 BMPER:412 TBX18:437 MKS1:1530 TXNRD1:1682 GFRF3:6080
Dementia	0.04602404	302	6.351e-03	8.476e-01	TS2H1:66 TGFBI:89 SORL1:189 PTK2B:193 BIN1:207 FZRL1:217 PRKN:290
ECG abnormality	-0.10316780	60	5.780e-03	8.476e-01	CPT2:9 PHYH:71 PEK6:88 SLC19A2:236 SYNE2:296 LARS2:390 TS2R:402
Enlarged kidney	-0.21764437	15	3.526e-03	8.476e-01	CPT2:9 PKHD1:159 NEK8:245 BMPER:412 FAH:1013 GPC3:1386 GPC6:1479
Epistaxis	-0.13112685	42	3.308e-03	8.476e-01	PTPN22:180 MPL:258 NBAL2:519 F7:718 ITGA2B:805 TET2:902 SLFN1A:906

KeggReactome Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_COMPLEX_I_BIOGENESIS	-0.30285285	48	4.214e-13	7.598e-10	NDUF6B:24 TMEM126B:86 NDUF3:127 NDUF9A:180 ECST:235 TMMD01:359 NDUF9A:8
REACTOME_NEURONAL_SYSTEM	0.10341376	352	4.914e-11	2.024e-08	KCNQ3:13 GABRB2:27 CACNA1B:30 ADCY1:38 DBNL5:4 SHANK2:59 KCNN1:17